

From Energy Observations to Continuum Error Certificates for Finite Fourier–Galerkin Navier–Stokes

Yaoharee Lahtee

Open Civil Science Initiative, Bangkok, Thailand

September 2026

Abstract

Two different completeness questions arise in finite numerical fluid dynamics. The first is *inner*: whether a declared observation process contains enough information to determine the retained finite state, up to unavoidable symmetry. The second is *outer*: whether the unrepresented continuum tail is provably small. A companion finite observability analysis of periodic three-dimensional Fourier–Galerkin Navier–Stokes gives all-resolution structural rank ceilings for total-energy and shell-energy Lie jets, positive-viscosity rank universality, and exact ceiling saturation in four certified reader–resolution cases. Separately, the Discrete Epsilon-Completion (EPSC) programme supplies target-indexed omitted-tail certificates, including Leray–Hopf spacetime bounds and an a posteriori residual-based terminal adapter with an exact-dyadic continuous-time reconstruction of recorded RK4 nodes. This paper composes the two layers. If an observation/inversion layer supplies a certified retained-state reconstruction radius ρ_N modulo an isometric symmetry preserving the Fourier cutoff, and EPSC supplies a certified omitted tail radius β_N , orthogonality gives

$$\inf_{g \in G} \|u(T) - g\hat{x}_N\|_2 \leq \sqrt{\rho_N^2 + \beta_N^2}.$$

The result makes precise the distinction between local observability rank and a quantitative measurement-to-continuum certificate. The missing inner-side ingredient is a certified inversion radius from energy jets; stable recovery under noisy high-order time information is a further open problem. These frontiers are distinct from the EPSC high-cutoff tightness/scalability problem. No claim about global regularity, finite-time singularity, global injectivity, or the Clay Millennium problem is made.

1 Introduction

A numerical state may be incomplete for two logically independent reasons. First, the sensors or readouts may not determine the finite state that the solver actually retains. Second, even a fully known finite state may omit continuum information above the cutoff. Treating these as one problem obscures both the mathematics and the engineering design space.

We therefore write the certification chain as

measurements $\longrightarrow$ retained-state certificate $\longrightarrow$ continuum-tail certificate $\longrightarrow$ final tolerance verdict.

The first arrow is an observability/inversion problem. The second is an EPSC problem. The present paper states their exact composition rule and identifies which parts are already established and which remain open.

The companion manuscript *Energy Observability Across Fourier–Galerkin Resolutions of Three-Dimensional Navier–Stokes* develops the finite observability side. The EPSC manuscript *Finite Readout Certificates for Navier–Stokes* develops the finite-to-continuum side. The purpose here is synthesis, not duplication.

2 Finite Fourier–Galerkin state and energy readers

For a cubic cutoff

$$K_N = \{-N, \dots, N\}^3 \setminus \{0\},$$

the declared real phase-space dimension is

$$\boxed{d_N = 2((2N + 1)^3 - 1)}. \quad (1)$$

Let $Q_N^{\text{shell}} = \{|k|^2 : k \in K_N\}$ and $m_N = |Q_N^{\text{shell}}|$. Let I be the vector of shell energies and $E = \mathbf{1}^T I$ the total retained kinetic energy. For the finite vector field F_ν , define the scalar and shell Lie jets

$$\mathcal{E}_R = (E, \mathcal{L}_F E, \dots, \mathcal{L}_F^R E), \quad \mathcal{J}_R = (I, \mathcal{L}_F I, \dots, \mathcal{L}_F^R I).$$

Spatial translation acts on Fourier coefficients by

$$(T_a \hat{u})_k = e^{ik \cdot a} \hat{u}_k.$$

The dynamics is translation-equivariant while both energy readers are invariant. At generic states where the translation action is locally free, no invariant observation map can exceed rank $d_N - 3$.

The finite observability analysis gives the structural ceilings

$$\boxed{\text{rank } D\mathcal{E}_R \leq \min(R + 1, d_N - 3)}, \quad (2)$$

$$\boxed{\text{rank } D\mathcal{J}_R \leq \min(m_N + (m_N - 1)R, d_N - 3)}. \quad (3)$$

Hence the symmetry ceiling cannot be attained before

$$R_E^{\min} = d_N - 4, \quad R_I^{\min} = \left\lceil \frac{d_N - 3 - m_N}{m_N - 1} \right\rceil. \quad (4)$$

For every positive viscosity, the scaling $x = \nu y$, $s = \nu t$ gives

$$\mathcal{L}_{F_\nu}^r h(\nu y) = \nu^{r+2} \mathcal{L}_{F_1}^r h(y), \quad D_x \mathcal{L}_{F_\nu}^r h(\nu y) = \nu^{r+1} D_y \mathcal{L}_{F_1}^r h(y),$$

for the quadratic energy readers. The resulting jet ranks are therefore independent of the value of $\nu > 0$. The Euler endpoint $\nu = 0$ is separate.

3 What is actually certified on the observability side

Exact modular certificates currently attain the translation ceiling at the earliest structurally admissible order in the following four cases:

N	reader	m_N	R	certified rank
1	total energy	1	48	49
1	shell energies	3	23	49
2	total energy	1	244	245
3	shell energies	18	39	681

Every row reaches $d_N - 3$. At a generic free-action state, this means that the kernel of the saturated jet differential is exactly the tangent space of the translation orbit. It is therefore appropriate to call the result *local finite-state completeness modulo translations*.

However, three stronger statements do not follow from rank saturation: global injectivity of the energy jet; a constructive inverse with branch control; or a numerical reconstruction radius under measurement error. The all-resolution assertion that the ceilings are generically attained at the earliest admissible order remains a conjecture.

The structural formulas also expose a measurement–time tradeoff. One scalar channel requires at least $d_N - 4 = \Theta(N^3)$ time derivatives before saturation is even possible. A multi-shell reader can reduce this depth. At $N = 3$, the scalar lower bound is $R \geq 680$, whereas the 18-channel shell reader is exactly saturated at $R = 39$. This is a structural observability statement, not yet a noise-robust sensor theorem.

4 The EPSC outer-completeness layer

Let P_N be the retained Fourier projection and $Q_N = I - P_N$. EPSC distinguishes nested finite agreement from a proved omitted-tail bound. A target-indexed certificate has the form

$$\|Q_N u\|_Y \leq \beta_{N,Y}.$$

For an unforced Leray–Hopf solution on the 2π -periodic three-torus,

$$\boxed{\|Q_N u\|_{L^2(0,T;L_x^2)} \leq \frac{\|u_0\|_2}{\sqrt{2\nu}(N+1)}}. \quad (5)$$

This is a genuine $O(N^{-1})$ spacetime tail certificate and does not assume global smoothness.

For a prescribed terminal time, richer retained information can yield a terminal bound. One route is the energy/dissipation budget; another uses a comparison path v and the residual

$$r = \partial_t v + \mathbb{P}[(v \cdot \nabla)v] - \nu \Delta v - \mathbb{P}f.$$

With

$$A_T = 2 \int_0^T \|\nabla v\|_\infty dt, \quad B_T = \int_0^T \|r\|_{H^{-1}}^2 dt,$$

the standard relative-energy estimate gives

$$\sup_{t \leq T} \|u(t) - v(t)\|_2^2 \leq e^{A_T} \left(e_0^2 + \frac{B_T}{\nu} \right). \quad (6)$$

If $v(T)$ is P_N -supported, the same bound controls the terminal omitted tail.

The current EPSC implementation supplies a concrete continuous comparison path from recorded RK4 nodes: every binary64 component is interpreted as its exact dyadic rational, the nodes are

projected exactly onto divergence-free Fourier space, and consecutive nodes are joined by a rational piecewise-linear path. The resulting Navier–Stokes residual is polynomial on each time cell, so its H^{-1} squared integral can be evaluated exactly in rational arithmetic. This closes the finite-tape enclosure construction registered as PROP-EPSC-15. Tightness and computational scaling at large cutoff/horizon remain the distinct open PROP-EPSC-16.

5 Observable-to-continuum composition

The two layers can now be connected without identifying local rank with continuum accuracy.

Theorem 1 (Observable-to-continuum orthogonal composition). *Let G be an isometric symmetry group on L^2 that preserves P_N . Suppose a certified finite observation/inversion procedure returns a P_N -supported state $\hat{x}_N$ and radius ρ_N such that*

$$\inf_{g \in G} \|P_N u(T) - g \hat{x}_N\|_2 \leq \rho_N. \quad (7)$$

Suppose independently that an EPSC argument supplies

$$\|Q_N u(T)\|_2 \leq \beta_N. \quad (8)$$

Then

$$\boxed{\inf_{g \in G} \|u(T) - g \hat{x}_N\|_2 \leq \sqrt{\rho_N^2 + \beta_N^2}.} \quad (9)$$

Proof. For any $g \in G$ admissible in (7), the difference $P_N u(T) - g \hat{x}_N$ lies in the retained subspace, while $Q_N u(T)$ lies in its orthogonal complement. Because $u = P_N u + Q_N u$,

$$\|u(T) - g \hat{x}_N\|_2^2 = \|P_N u(T) - g \hat{x}_N\|_2^2 + \|Q_N u(T)\|_2^2.$$

Apply the two certified bounds and take the infimum over g . □

For the energy-observability application, $G = \mathbb{T}^3$ acts by spatial translations. Thus a final fail-closed tolerance rule is

$$\boxed{\sqrt{\rho_N^2 + \beta_N^2} \leq \varepsilon \implies \text{continuum state certified modulo translation at tolerance } \varepsilon.} \quad (10)$$

If either ρ_N or β_N is absent, the verdict is **HOLD**.

Remark 2 (Why this is sharper than addition). A generic triangle inequality would give $\rho_N + \beta_N$. The Fourier split is orthogonal, so the retained and omitted errors combine Pythagoreanly. This is useful because the observation-side and tail-side budgets can be allocated independently under a single final tolerance.

6 Symmetry compatibility

Total and shell energies are invariant under spatial translations, so finite observability can at best recover a translation orbit. This ambiguity is not a defect of the reader; it is the symmetry of the model. The continuum L^2 tail norm is also translation invariant, and P_N commutes with translations. Therefore Theorem 1 naturally lives on the quotient by translations.

For residual-based EPSC certificates, the exact mathematical quantities A_T and B_T are translation invariant when the comparison path and solution are translated together. A specific computer majorant may nevertheless depend on the chosen Fourier phase representation and can be looser for one representative than another. Certificate validity is preserved; tightness and representative optimisation belong to the engineering/scalability frontier rather than to the observability theorem.

7 The three frontiers after composition

The combined architecture separates three open tasks that should not be collapsed into one.

F1: all-resolution observability saturation. The structural ceilings (2)–(3) hold at every finite resolution, but exact generic saturation at the earliest admissible depth is certified only in the four recorded cases. The all- N statement remains PROP-NSOBS-07.

F2: certified finite inversion and noise stability. Rank saturation yields local differential identifiability modulo translations, but the composition theorem needs a certified radius ρ_N . Constructing such an inverse is PROP-EPSC-18; propagating measurement and differentiation uncertainty through it is PROP-EPSC-19.

F3: scalable outer certification. The exact-dyadic EPSC-15 path construction is a correctness route, but its conservative gradient majorants and residual computation may become expensive or loose as N and T grow. Improving that cost/tightness is PROP-EPSC-16.

These tasks interact in an eventual practical system, but they are mathematically different.

8 Executable evidence and provenance

The repository reproduction ledger contains finite checks for the observability bookkeeping, recorded saturation ceilings, translation invariance of the energy readers, the orthogonal $\sqrt{\rho^2 + \beta^2}$ composition, EPSC finite algebra, and the exact-dyadic continuous-time RK4 path enclosure. Standard analytic implications remain labelled at the analytic/derived tier rather than being presented as machine proofs.

Proposal provenance is split deliberately. The NS observability family is registered as PROP-NSOBS-01 through PROP-NSOBS-08. The EPSC composition extension is registered as PROP-EPSC-17 through PROP-EPSC-19; PROP-EPSC-16 remains the separate high-cutoff tightness/scalability frontier. Proposal identifiers are not automatically canonical Toledo theorem codes.

9 Claim boundary

The strongest new statement here is a conditional composition theorem: a certified finite reconstruction radius and a certified omitted-tail radius combine orthogonally into a continuum quotient-state radius. This does not turn qualitative local rank into a numerical inverse, and it does not prove that total or shell energy observations are globally injective. The exact finite saturation results remain finite-resolution observability results. The EPSC certificates remain target- and assumption-specific.

Nothing in this paper proves global smoothness, finite-time blow-up, uniqueness of all Leray–Hopf weak solutions, continuum observability from finitely many noisy sensors, turbulent DNS adequacy at coarse cutoff, or the Clay Millennium existence-and-smoothness problem.

10 Conclusion

Energy observability and EPSC close opposite sides of the same measurement-to-continuum pipeline. The former asks whether the represented state is visible; the latter asks how much unrepresented state can remain. Their rigorous interface is not “rank plus convergence looks good”, but two certified

radii. Once a finite inversion radius ρ_N and an omitted-tail radius β_N are available, the final L^2 quotient-state certificate is

$$\sqrt{\rho_N^2 + \beta_N^2}.$$

This cleanly separates what has been established from what remains: all-resolution saturation, quantitative/noise-stable energy-jet inversion, and scalable tight outer certification.

References

- [1] J. Leray, Sur le mouvement d’un liquide visqueux emplissant l’espace, *Acta Mathematica* 63 (1934), 193–248.
- [2] E. Hopf, Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen, *Mathematische Nachrichten* 4 (1951), 213–231.
- [3] R. Temam, *Navier–Stokes Equations: Theory and Numerical Analysis*, AMS Chelsea Publishing.
- [4] C. L. Fefferman, Existence and smoothness of the Navier–Stokes equation, Clay Mathematics Institute Millennium Problem description, 2000.
- [5] Y. Lahtee, A Readout Problem for Fefferman’s Existence and Smoothness of the Navier–Stokes Equation, Zenodo, DOI: 10.5281/zenodo.22673246, 2026.
- [6] Y. Lahtee, Energy Observability Across Fourier–Galerkin Resolutions of Three-Dimensional Navier–Stokes, companion manuscript and executable record, 2026.
- [7] Y. Lahtee, Finite Readout Certificates for Navier–Stokes: Discrete Epsilon-Completion, Spectral Tails, and A Posteriori Adapters, companion manuscript and executable record, 2026.